

Which DSA pattern? — a decision guide

Is the array sorted (or sortable)?

Yes → **Two Pointers / Binary Search**

- Subarray / substring window? → **Sliding Window**
- Linked list? → **Fast & Slow / Reversal**
- Shortest path / min steps? → **BFS**
- All paths / components? → **DFS / Backtracking**
- Dependency order? → **Topological Sort**
- Dynamic connectivity? → **Union-Find**
- Top-K / K-way merge? → **Heap**
- Overlapping intervals? → **Merge Intervals**
- Count ways / optimal value? → **Dynamic Programming**
- Prefix / autocomplete? → **Trie**
- Next greater / histogram? → **Monotonic Stack**
- Subarray-sum queries? → **Prefix Sum**